23AID498 – Project Phase 1

Project Diary

Student Name:	Sheron George
Roll No:	AM.AI.U4AID23059
Team No:	4
Guide Name:	Dr. Divya Udayan J
Project Title:	Optimizing Autonomous Traffic Navigation via Reinforcement Learning and Basal Ganglia Gating in ROS2

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 14/07/26

Work planned for the coming week	• Identify and instantiate ego vehicle as the central authonomous agent within the SUMO network. • Implement a bidirectional communication link between simulation environment and controller • Setup motor control feedback loops & spatial tracking mechanisms.
Work completed during the previous week	• Integrated SUMO envt. • Setup the periodic traffic light controller • Setup the SUMO vehicle control • Setup bidirectional road lane
Pending work	• Need to setup PPO policy action • Need to integrate a monitoring system to detect the traffic light phases

Remarks by the Guide:

Progressing.

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date:

Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 16/07/26

Work planned for the coming week	• Need to enable hippocampal module to map vehicle's position. • Need to integrate a safety critical control
Work completed during the previous week	• Literature review → Collected 10 journals and compared with our project → Followed by novelty identification and phase I, II division → Set motor control
Pending work	

Remarks by the Guide:

Progressing.

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date:

Dr. Divya Udayan J

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 22/07/26

Work planned for the coming week	• Planning to make the model dynamic • Increase the complexity of the model
Work completed during the previous week	• Enable hippocampal module to map vehicle's position • Integrated a safety critical control • Drafted Literature - PP I
Pending work	

Remarks by the Guide: ppt prescated.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: 22/7/26

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 30/7/2026

Work planned for the coming week	• Increased the time limit of model execution • Improve the Action Selection process by adding SAC
Work completed during the previous week	• Increased the complexity of model by increasing traffic and adjusted the signal simulation • Drafted System Design
Pending work	

Remarks by the Guide: Progressing

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date:

Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 07/08/26

Work planned for the coming week	• Execute long horizon PPO training sessions within the newly upgraded dynamic environment • Analyze and plot rewards convergence graphs from the new training runs • Quantify the agent's robustness by evaluating collision rates and trajectory smoothness against high sensor noise.
Work completed during the previous week	• Finalized PPO integration within the bio-inspired Hippocampal-Basal Ganglia framework. • Programmed specialized multi-objective reward shaping • Injected Gaussian noise into kinematic observation to emulate realistic ROS2 sensor inaccuracies
Pending work	• Overhauled the evaluation pipeline to bypass step limits and support infinite continuous GUI rendering

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 02/09/2026

Work planned for the coming week	• Upgrade the model setup from standard SAC to hybrid SAC • Improve policy's robustness for chaotic and high traffic environments • Begin preparing the training policy for integration with ROS 2
Work completed during the previous week	• Completed SAC setup and training • Evaluated current policy in preparation for transition to hybrid SAC • Identified requirements for ROS 2 integration and /cmd_vel command
Pending work	• Complete the transition to the hybrid SAC configuration • Continue policy training and evaluation under chaotic traffic • Connect trained policy to ROS 2 and begin /cmd_vel transmission testing.

Remarks by the Guide: Progressing

4.5

Weekly Performance Rating (0–5): ____ / 5

04/9/25

Dr. Divya Udayan

Guide's Signature with Date:

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 09/09/2026

Work planned for the coming week	• Connect trained SAC policy to ROS 2 • Test /cmd_vel pipeline and verify velocity commands are transmitted correctly • Run the traffic density evaluations required for RQ3 • Begin final preparations for hardware
Work completed during the previous week	• Upgraded model from SAC to hybrid SAC • Continued training and evaluation of hybrid SAC policy for improved robustness in chaotic traffic zones • Prepared policy and integration components required for ROS 2 testing
Pending work	• Complete ROS 2 integration and validate /cmd_vel communication • Finish the RQ3 traffic density evaluation • Finalize the hardware deployment setup • Conduct physical testing and validate the system on the hardware

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: 09/9/25

Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 16/9/26

Work planned for the coming week	- Evaluated and benchmark the proposed TD3+Bio-shield architecture - Generate data visualizations, including the cumulative episode rewards progression plots and empirical ablation bar charts
Work completed during the previous week	- Implemented and trained unshielded SAC and PPO baseline algorithms for traffic simulation - Assisted in debugging the TraCI action space mapping, ensuring agent's continuous [-1.0, 1.0] throttle/brake outputs translated correctly to ego vehicle's physics
Pending work	- Conducting the initial evaluation runs for the TD3+Bio-shield architecture - Simulating data for graphical planning

Remarks by the Guide: Code review discussed.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divyabharathi

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 24/9/26

Work planned for the coming week	• Contributed to drafting the final codebase, model checkpoints and documentation for upcoming project handoff • Execute final robustness stress-tests on the trained TD3 model specifically isolating high density urban traffic scenarios. • Assist in finalizing the project manuscript.
Work completed during the previous week	• Evaluated and benchmarked the proposed TD3 + Bio-Shield architecture against the baselines, logging mean episode rewards, collision rates and intervention frequencies • Generated and refined the data visualizations, including the cumulative episode reward progression plots and 3-panel empirical bar charts
Pending work	• Final verification of the logged benchmark data against the LaTeX tables and PDF plots to guarantee exact metric alignment • Archiving and documenting the trained model weights and hyperparameter configurations

Remarks by the Guide: Progressing

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 08/10/2026

Work planned for the coming week	• Implement, train and benchmark the unshielded Deep Deterministic Policy Gradient algorithm to expand the comparative analysis against the existing SAC, PPO and TD3 models
Work completed during the previous week	• Executed final robustness stress-tests on the trained TD3 model, specifically isolating and evaluating high density urban traffic scenarios • Code review completed • Packaged the final codebase, model checkpoints, technical documentation for project
Pending work	• Finalizing long term storage of raw simulation data logs and pretrained model weights

Remarks by the Guide: Progressing

4.5

Weekly Performance Rating (0–5): ____ / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 13/10/26

Work planned for the coming week	• Execute target policy smoothing experiments on DDPG to directly quantify how much value overestimation affects braking distances comparing to TD3 • Run hyperparameter sweeps over actor critic rates and soft update factors to eliminate reward collapse in traffic
Work completed during the previous week	• Built and trained DDPG baseline networks, configuring dual continuous actor and critical networks with target n/w polyak averaging (τ= 0.005) • Implemented replay buffer mechanics and stabilized the deterministic policy gradient updates against sparse terminal collision states
Pending work	• Profiling critic loss variance under high-density vehicle spawning to identify policy divergence policy. • Structuring raw elements and logs into numerical matrix exports for ablation analysis

Remarks by the Guide: Dr. Divyalbalayan (Progressing)

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divyalbalayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 23/10/2025

Work planned for the coming week	Need to write the report Implementation of comparison study of all the models.
Work completed during the previous week	Implemented, trained and benchmarked the DDPG algorithm to expand, analysis. Finalized the conference paper
Pending work	

Remarks by the Guide: Progressing.

4.5

Weekly Performance Rating (0–5): ____ / 5

Guide's Signature with Date: Dr. Divya Udayan